Ice Rheology: Creep Mechanisms and Glen's Flow Law

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Topics Covered

- The four creep mechanisms in polycrystalline ice
- The composite flow law of Goldsby & Kohlstedt (2001)
- How the composite flow law reduces to Glen's flow law
- Physical basis for the rate factor A and the creep exponent n

Primary reference: Cuffey & Paterson, *Physics of Glaciers*, 4th ed., Chapter 3; Goldsby & Kohlstedt (2001).

1 Introduction and Motivation

In the preceding lectures, we stated Glen's flow law as the constitutive relation for glacier ice:

$$\dot{\varepsilon}_{ij} = A\,\tau_e^{n-1}\,\tau_{ij}, \tag{1}$$

with $n = 3$ the typical assumption and A a temperature-dependent rate factor. We used this relation to derive velocity profiles, build the SIA, formulate the full Stokes system, and study the SSA equations. Throughout, we treated A and n as given without examining their physical basis.

The goal of this lecture is to understand where Eq. 1 comes from.

Ice is a polycrystalline material, and its deformation is governed by several grain-scale mechanisms that operate in parallel or in series depending on the stress, temperature, and grain size. A grain is a single crystal and the terms 'grain' and 'crystal' are often casually interchanged. Polycrystalline simply means there are many grains. Understanding the deformation mechanisms is important for at least three reasons:

1. It reveals when Glen's law is a good approximation and when it breaks down.
2. It explains the temperature dependence of the rate factor A through activation energies.
3. It clarifies the physical meaning of the creep exponent n and why the commonly assumed value of $n = 3$ is not a material property but rather a special case that, unfortunately for the field, was casually taken to be general several decades ago. The near-universal assumption of $n = 3$ has severely limited progress in glacier dynamics, as will no doubt become increasingly clear in the coming years and decades.

Plan for this lecture. We first review the general form of creep flow laws for crystalline materials (§2). We then discuss the four creep mechanisms identified in ice (§3). Next, we present the composite flow law that combines all four mechanisms (§4). Finally, we show how this composite law reduces to Glen's flow law under the conditions typically assumed in glaciology (§6).

2 General Form of Creep Flow Laws

Rheological data for crystalline materials are typically analyzed with a power law of the form

$$\boxed{\dot{\varepsilon}_e = A_0 \frac{\tau_e^n}{d^r} \exp\left(-\frac{Q + pV}{RT}\right),} \tag{2}$$

where:

- $\dot{\varepsilon}_e$ is the effective strain rate and τ_e is the effective deviatoric stress,
- A_0 is a material-dependent prefactor,
- n is the stress exponent,
- d is the mean grain size and r is the grain-size exponent,
- Q is the activation energy for creep,
- p is the pressure and V is the activation volume,
- R is the gas constant and T is the absolute temperature.

The values of n, r, and Q are diagnostic: they identify which physical mechanism controls the creep rate. For example, a stress exponent $n = 1$ indicates diffusion-controlled flow, while $n > 1$ indicates a dislocation-mediated process. A grain-size exponent $r > 0$ indicates that grain boundaries participate in the rate-limiting process.

The activation volume V describes the pressure dependence of the creep rate. For ice, $V \approx -13 \times 10^{-6}$ $\mathrm{m^3\,mol^{-1}}$ [1]. The pV term modifies the effective activation energy under pressure: at the base of a 3 km-thick ice sheet, $p \approx \rho_i g h \approx 27$ MPa, giving $pV \approx -0.35\ \mathrm{kJ\,mol^{-1}}$, which is negligible compared to $Q \approx 60\ \mathrm{kJ\,mol^{-1}}$. The activation volume is therefore typically omitted in glaciological applications, reducing the Arrhenius factor to $\exp(-Q/RT)$.

3 The Four Creep Mechanisms in Ice

Laboratory experiments on fine-grained ice by [1] reveal four distinct creep regimes, each characterized by different values of n and r. A useful organizing principle is to distinguish between **intragranular** deformation mechanisms and **intergranular** accommodation mechanisms:

- **Intragranular** mechanisms deform the interior of individual grains. In ice, these are dislocation creep (the motion of dislocations through the crystal lattice) and diffusion (the transport of molecules through the lattice or along grain boundaries driven by stress gradients). These are the fundamental crystalline deformation mechanisms—they produce strain.
- **Intergranular** accommodation mechanisms operate at grain boundaries to maintain compatibility between neighboring grains. The most important is grain boundary sliding (GBS), in which neighboring grains translate and rotate relative to one another along their shared boundary.

A single crystal deforming in isolation needs only intragranular mechanisms. In a polycrystal, however, grains must deform compatibly with their neighbors—each grain is constrained by the grains surrounding it. This compatibility requirement means that the intragranular deformation mechanisms must be accompanied by an accommodation mechanism that resolves the mismatch at grain boundaries. Accommodation can be achieved either intragranularly (by activating hard slip systems within each grain, as in dislocation creep) or intergranularly (by GBS). The nature of this coupling—which mechanism is rate-limiting—determines the observed creep parameters.

The four creep regimes identified by [1] are, in order of decreasing stress: dislocation creep, GBS-accommodated basal slip ("superplastic flow"), basal-slip-accommodated GBS, and diffusional flow. Each is described by a flow law of the form given in equation (2). We discuss each mechanism below.

3.1 Dislocation Creep ($n = 4.0$, $r = 0$)

At the highest stresses, deformation is controlled by the motion of dislocations through the crystal lattice—a purely intragranular process. Dislocations are line defects in the crystal structure, and their motion under an applied stress produces macroscopic strain. In ice, the dominant slip system is the basal plane (the hexagonal plane perpendicular to the c-axis), on which dislocations glide easily. Non-basal slip (on prismatic and pyramidal planes) is much more difficult—single crystals oriented for easy basal slip can deform $> 10^5$ times faster than those oriented for hard slip.

Dislocation creep in polycrystalline ice is characterized by:

- A stress exponent $n = 4.0$, measured consistently in experiments at high confining pressure.
- No grain-size dependence ($r = 0$): the strain rate depends only on stress and temperature, because the rate-limiting step is dislocation motion within grains.
- An activation energy $Q \approx 60$ kJ mol^{-1} below ~258 K, increasing to ~180 kJ mol^{-1} above 258 K due to premelting at grain boundaries.

The rate-limiting step in dislocation creep is dislocation climb, in which edge dislocations move perpendicular to their glide plane by absorbing or emitting point defects. Climb allows dislocations to bypass obstacles (other dislocations, grain boundaries), and is controlled by volume self-diffusion. The equivalence of the activation energy for dislocation creep (~60 kJ mol^{-1}) with that for volume self-diffusion of hydrogen and oxygen in ice confirms that climb limits the creep rate.

The creep strength of polycrystalline ice in the dislocation creep regime is intermediate between the strengths of single crystals oriented for easy (basal) and hard (non-basal) slip. This reflects the fact that each grain in a polycrystal must deform compatibly with its neighbors, requiring activation of both basal and non-basal slip systems. In this regime, the intergranular accommodation is achieved entirely by intragranular means—hard (non-basal) slip within each grain resolves the mismatch at grain boundaries, so GBS plays no role and $r = 0$.

3.2 GBS-Accommodated Basal Slip — Superplastic Flow ($n = 1.8$, $r = 1.4$)

At intermediate stresses, a transition occurs to a creep regime characterized by $n = 1.8$ and a strong dependence on grain size ($r = 1.4$). In this regime, the intragranular mechanism is basal slip (easy glide on the basal plane), while the intergranular accommodation mechanism is grain boundary sliding (GBS).

Recall that in dislocation creep, grain-to-grain compatibility is achieved by an intragranular process: activation of hard (non-basal) slip systems within each grain. In the superplastic regime, an intergranular process takes over this role: grains slide past one another along their boundaries, accommodating the mismatch produced by easy basal slip within each grain. Because GBS is the slower of the two serial processes (basal slip and GBS) under these conditions, GBS limits the overall strain rate.

Key point: serial vs. parallel processes

When two deformation processes operate in **series** (both must occur for macroscopic strain), the slower process limits the rate:

$$\dot{\varepsilon}_{\text{total}}^{-1} = \dot{\varepsilon}_{\text{basal}}^{-1} + \dot{\varepsilon}_{\text{GBS}}^{-1}. \tag{3}$$

When two processes operate in **parallel** (either alone can produce macroscopic strain), their rates add:

$$\dot{\varepsilon}_{\text{total}} = \dot{\varepsilon}_1 + \dot{\varepsilon}_2. \tag{4}$$

Basal slip (intragranular) and GBS (intergranular) are serial processes: basal slip produces strain within grains, but without GBS to accommodate grain-to-grain incompatibility, the polycrystal cannot deform. The transition from dislocation creep ($n = 4$) to superplastic flow ($n = 1.8$) occurs when GBS becomes slower than the hard-slip accommodation mechanism and thereby becomes the rate-limiting step. This is a transition in the intergranular accommodation mechanism—from hard slip within grains to sliding between grains.

The superplastic regime is characterized by:

- A stress exponent $n = 1.8$.
- A grain-size exponent $r = 1.4$: finer-grained ice deforms faster because there is more grain-boundary area per unit volume.
- An activation energy $Q \approx 49$ kJ mol^{-1} below $\sim$258 K, increasing at higher temperatures due to premelting.

Microstructural evidence supports the role of GBS in this regime. Samples deformed in the $n = 1.8$ regime show minimal grain flattening (unlike dislocation creep, where grains are substantially deformed), numerous four-grain junctions (a signature of grain switching associated with GBS), and straight, regular grain boundaries (unlike the irregular boundaries produced by dynamic recrystallization in dislocation creep).

3.3 Basal-Slip-Accommodated GBS ($n = 2.4$, $r = 0$)

At still lower stresses and for the finest grain sizes ($d \lesssim 5$ μm), a third creep regime appears with $n = 2.4$ and no grain-size dependence ($r = 0$). Here, the intergranular mechanism (GBS) is faster than the intragranular mechanism (basal slip), so basal slip becomes the rate-limiting step.

This represents the other limit of the serial intragranular/intergranular process. In the superplastic regime ($n = 1.8$), the intergranular process (GBS) is slower and controls the rate. At lower stresses, the strain-rate contributions cross over because of their different stress exponents, and now the intragranular process (basal slip) is the slower of the two. Because the rate-limiting step is intragranular, the creep rate is independent of grain size ($r = 0$).

The creep parameters in this regime—$n = 2.4$ and $Q \approx 60$ kJ mol^{-1}—are in excellent agreement with those for basal slip in single crystals, confirming that basal slip is rate-limiting. The absolute creep strengths of polycrystalline samples in this regime also agree well with single-crystal basal-slip data. This regime has only been observed experimentally for very fine-grained samples ($d \lesssim 5$ μm) at very low stresses and is likely inaccessible for the grain sizes relevant to glacier flow.

3.4 Diffusional Flow ($n = 1.0$, $r = 2$–3)

At the lowest stresses, deformation can in principle occur by diffusion of molecules either through the crystal lattice (Nabarro–Herring creep, $r = 2$) or along grain boundaries (Coble creep, $r = 3$). Both are intragranular deformation mechanisms—they produce strain by rearranging molecules within or between grains in response to stress gradients. The grain-size dependence ($r > 0$) arises because smaller grains have shorter diffusion paths, not because diffusion is an accommodation mechanism. Both are characterized by a stress exponent $n = 1$ (Newtonian behavior).

The diffusional flow rate for polycrystalline ice can be written as

$$\dot{\varepsilon}_{\text{diff}} = \frac{42\,\tau_e\,V_m}{R\,T\,d^2}\left(D_v + \frac{\pi\,\delta}{d}\,D_b\right), \tag{5}$$

where V_m is the molar volume, D_v is the volume diffusion coefficient, δ is the grain-boundary width, and D_b is the grain-boundary diffusion coefficient.

Based on estimates of the diffusion parameters for ice, [1] conclude that diffusional flow is unattainably slow at laboratory strain rates for grain sizes $\gtrsim 3$ μm. For glaciologically relevant grain sizes (typically ≥ 1 mm), the diffusional flow rate is negligible compared to the other creep mechanisms. Diffusional flow has not been observed experimentally in ice even for samples with grain sizes as small as 3 μm.

Key point: the dominant creep mechanism depends on conditions

The four creep mechanisms are summarized below. At glaciologically relevant conditions (shear stresses $\lesssim 0.1$ MPa, grain sizes ≥ 1 mm, temperatures 233–273 K), the superplastic flow regime ($n = 1.8$) is predicted to be the rate-limiting mechanism for much of the stress range encountered in glaciers and ice sheets.

Mechanism	n	r	Q (kJ mol^{-1})	**Scale**	**Rate-limiting step**
Dislocation creep	4.0	0	60	Intra	Dislocation climb
GBS-accomm. basal slip	1.8	1.4	49	Inter	Grain boundary sliding
Basal-slip-accomm. GBS	2.4	0	60	Intra	Basal slip
Diffusional flow	1.0	2–3	$\sim$60	Intra	Vol./boundary diffusion

4 The Composite Flow Law

The flow of polycrystalline ice cannot be adequately described by a single power law (i.e., Glen's Flow Law) with one set of creep parameters. [1] proposed a composite constitutive equation that combines all four creep mechanisms:

$$\boxed{\dot{\varepsilon}_e = \dot{\varepsilon}_{\text{diff}} + \left(\frac{1}{\dot{\varepsilon}_{\text{basal}}} + \frac{1}{\dot{\varepsilon}_{\text{gbs}}}\right)^{-1} + \dot{\varepsilon}_{\text{disl}},} \tag{6}$$

where $\dot{\varepsilon}_{\text{diff}}$ is the diffusional flow rate, $\dot{\varepsilon}_{\text{basal}}$ is the basal-slip rate, $\dot{\varepsilon}_{\text{gbs}}$ is the GBS rate, and $\dot{\varepsilon}_{\text{disl}}$ is the dislocation creep rate. It can be shown that this is the minimum viable representation for viscous (creep) deformation in isotropic ice Ih (all ice we're interested in on Earth is ice Ih).

The structure of equation (6) reflects the physical relationships between intragranular and intergranular processes:

- **Basal slip (intragranular) and GBS (intergranular) are in series** (the harmonic mean): both must operate for grain-size-sensitive deformation, and the slower process limits the rate. At high stresses, the intergranular process (GBS) is slower and the effective stress exponent approaches $n = 1.8$; at low stresses, the intragranular process (basal slip) is slower and $n \to 2.4$.
- **Dislocation creep (intragranular) operates in parallel** with the grain-size-sensitive mechanisms: it provides an independent intragranular deformation pathway through the activation of multiple slip systems, with hard slip serving as the intergranular accommodation mechanism. At high stresses, dislocation creep dominates because of its high stress exponent ($n = 4$).
- **Diffusional flow also operates in parallel**: it provides yet another independent intragranular deformation pathway, dominant only at very low stresses and very small grain sizes.

Each term in equation (6) has the form of equation (2), with parameters given in Table 1.

Table 1: Constitutive equation parameters from [1]. Two sets of parameters are given for dislocation creep and GBS-accommodated basal slip to account for the increase in activation energy above ~258 K due to premelting.

Creep regime	A_0 (units vary)	n	r	Q ($\mathrm{kJ\,mol^{-1}}$)
Dislocation creep ($T < 258$ K)	4.0×10^{5} $\mathrm{MPa^{-4.0}\,s^{-1}}$	4.0	0	60
Dislocation creep ($T > 258$ K)	6.0×10^{28} $\mathrm{MPa^{-4.0}\,s^{-1}}$	4.0	0	~181
GBS-accomm. basal slip ($T < 255$ K)	3.9×10^{-3} $\mathrm{MPa^{-1.8}\,m^{1.4}\,s^{-1}}$	1.8	1.4	49
GBS-accomm. basal slip ($T > 255$ K)	3.0×10^{26} $\mathrm{MPa^{-1.8}\,m^{1.4}\,s^{-1}}$	1.8	1.4	~192
Basal-slip-accomm. GBS	5.5×10^{7} $\mathrm{MPa^{-2.4}\,s^{-1}}$	2.4	0	60

Predictions of this composite flow law are in excellent agreement with published creep data for coarse-grained ($d \geq 1$ mm) polycrystalline ice across a wide range of temperatures and stresses.

5 Deformation Mechanism Maps

The composite flow law involves multiple mechanisms with different dependencies on stress, temperature, and grain size. A **deformation mechanism map** is a graphical tool that shows which mechanism dominates at each point in parameter space. These maps are widely used in materials science and provide an intuitive picture of the rheological behavior of a material across a range of conditions.

5.1 Strain Rate vs. Stress

The most direct view of the composite flow law is a log-log plot of $\dot{\varepsilon}_e$ vs. τ_e at fixed temperature and grain size (Figure 1). At high stresses, the composite curve follows the dislocation creep line (slope

= 4). At lower stresses, it transitions to the GBS-accommodated basal slip line (slope = 1.8). Glen's original experiments were conducted in the transition region between these two regimes, where the apparent slope is ∼3.

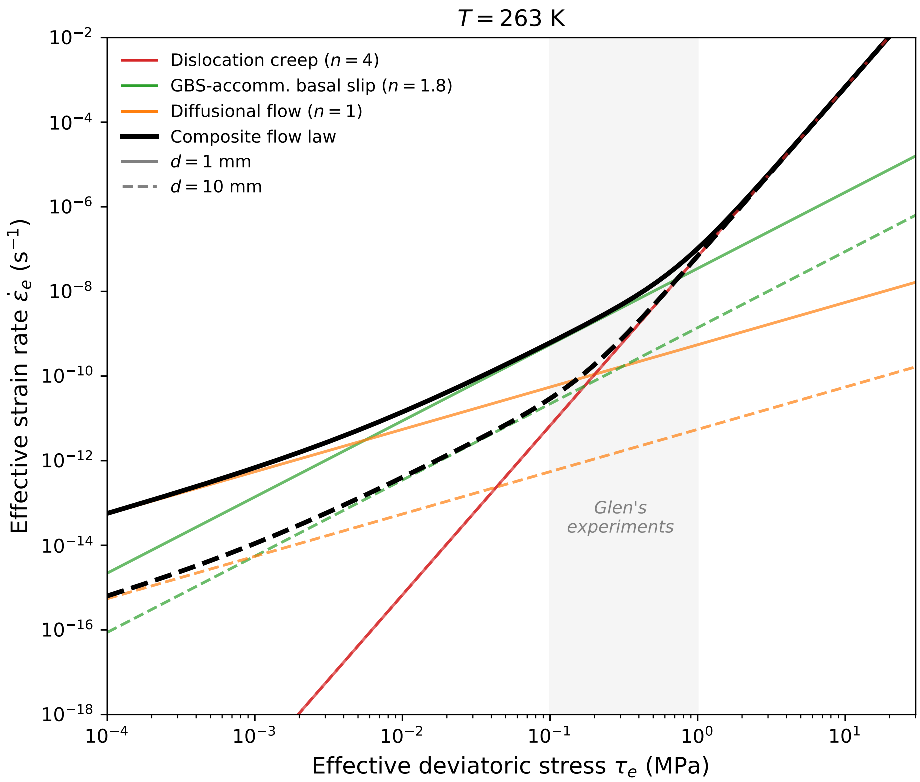

Figure 1: Log-log plot of effective strain rate vs. effective deviatoric stress at $T = 263$ K for two grain sizes: $d = 1$ mm (solid lines) and $d = 10$ mm (dashed lines). Colors indicate the creep mechanism; black curves show the composite flow law. Dislocation creep (red) is grain-size independent, so the solid and dashed red lines overlap. GBS-accommodated basal slip (green) and diffusional flow (orange) are grain-size sensitive: their rates decrease for larger grains. The crossover stress between GBS-dominated and dislocation-dominated flow shifts to lower values for larger grains. Glen's experimental stress range (0.1–1 MPa, shaded) straddles this transition.

5.2 Stress–Temperature Map

At fixed grain size, we can color the τ_e–T plane by the dominant mechanism (Figure 2). For stresses above ∼1 MPa, dislocation creep dominates at all temperatures. Below ∼0.1 MPa, superplastic flow dominates. The boundary between these regimes is the locus of the crossover stress τ_e^* discussed in §6, and it shifts with temperature because the two mechanisms have different activation energies.

5.3 Stress–Grain Size Map

At fixed temperature, the τ_e–d map (Figure 3) reveals the role of grain size. For large grains ($d \gtrsim 10$ mm), the grain-size-sensitive mechanisms (GBS-accommodated basal slip) require each grain boundary to accommodate more slip, reducing the GBS strain rate and expanding the domain

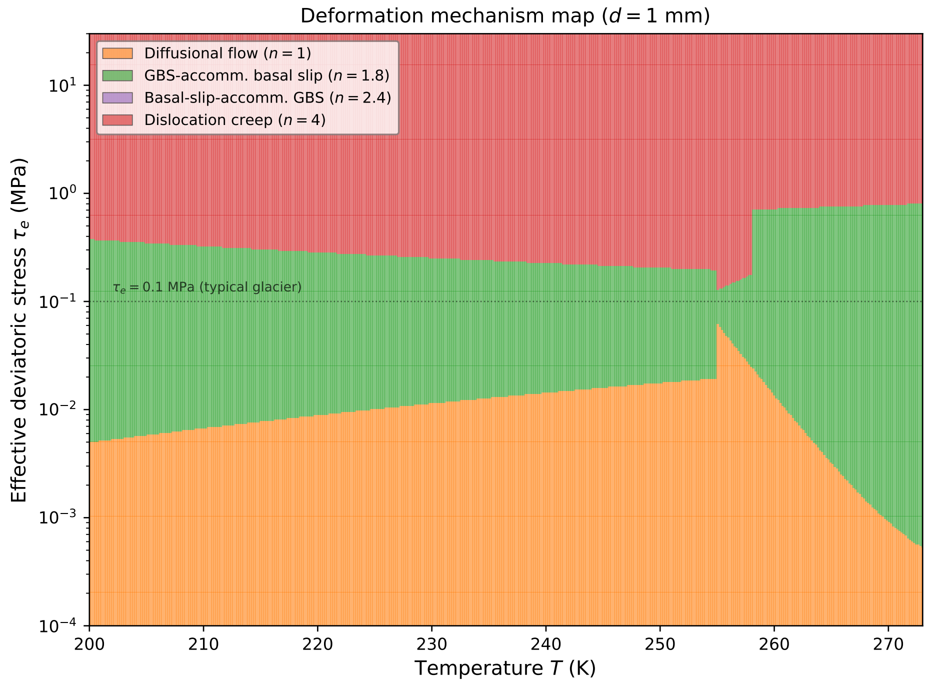

Figure 2: Deformation mechanism map in τ_e–T space for $d = 1$ mm. Colors indicate the dominant creep mechanism. The dashed line marks $\tau_e = 0.1$ MPa, a typical stress scale for glaciers. At glaciological stresses, superplastic flow ($n = 1.8$) dominates at low temperatures, while dislocation creep ($n = 4$) can become important near the melting point due to the increase in activation energy.

of dislocation creep to lower stresses. For small grains ($d \lesssim 1$ mm), the transition stress τ_e^* shifts upward and superplastic flow dominates over a wider stress range.

5.4 The Effective Stress Exponent

A useful complement to the mechanism map is a plot of the effective stress exponent n_{eff} as a function of stress (Figure 4). This makes concrete the idea that $n = 3$ is not a fundamental value but a transitional one.

5.5 Effect of Grain Size on the Composite Flow Law

Figure 5 shows the composite flow law for several grain sizes at fixed temperature. At high stresses, all curves converge to the dislocation creep line (which is grain-size independent). At lower stresses, finer-grained ice deforms faster because the GBS-accommodated basal slip mechanism ($r = 1.4$) is more efficient. The crossover stress between dislocation-dominated and GBS-dominated flow shifts to higher values for finer grains.

6 From the Composite Flow Law to Glen's Flow Law

In glaciology, the composite flow law (6) has traditionally been replaced by the much simpler Glen's flow law:

$$\dot{\varepsilon}_e = A\,\tau_e^{\,n}, \tag{7}$$

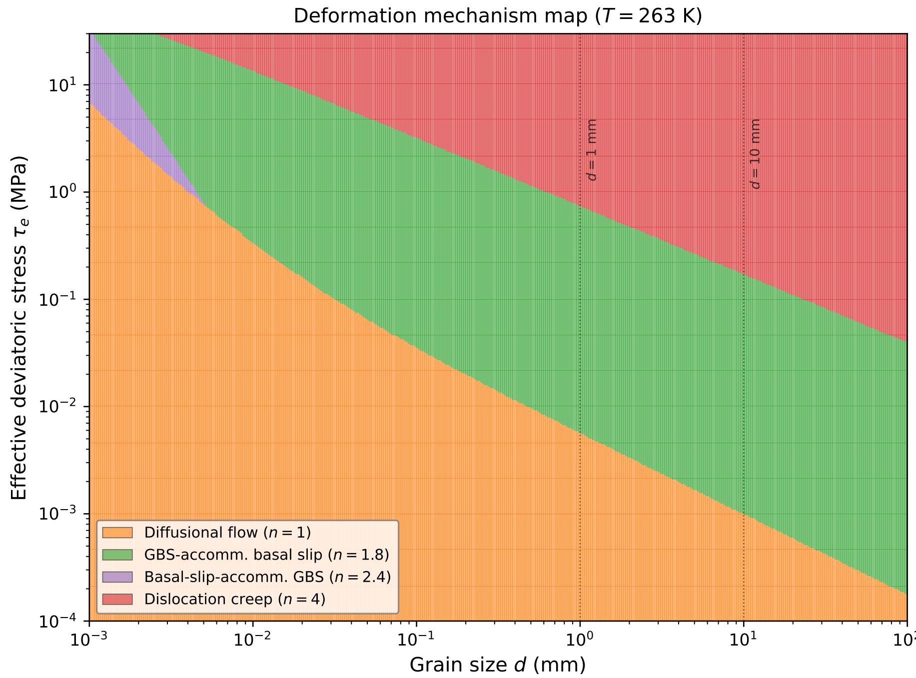

Figure 3: Deformation mechanism map in τ_e–d space for $T = 263$ K. Vertical lines mark $d = 1$ mm and $d = 10$ mm, spanning the range typical of glacier ice. Finer-grained ice (e.g., ice-age ice with high impurity content) has a broader superplastic flow domain.

where n is taken to be a constant (usually $n = 3$) and A absorbs all dependence on temperature and grain size. This simplification traces back to the pioneering experiments of [2, 3], conducted on polycrystalline ice at stresses of 0.1–1 MPa and temperatures near the melting point. Here we show how the composite flow law reduces to Glen's law under certain conditions, and what is lost in the reduction.

6.1 Glen's Original Experiments

Glen's experiments were performed on coarse-grained polycrystalline ice ($d \geq 1$ mm) at differential stresses of approximately 0.1–1 MPa and temperatures near the melting point. His data yielded stress exponents ranging from roughly 3 to 4, depending on the stress range considered. It was Nye [9] and subsequent workers who adopted the single value $n = 3$ as a convenient simplification for analytical glacier-flow models. As shown in Figure 1 of [1], Glen's data fall precisely in the transition region between the dislocation creep regime ($n = 4$) and the superplastic flow regime ($n = 1.8$).

Fitting a single power law through data that span a transition between two regimes yields an *apparent* stress exponent that is intermediate between the exponents of the two regimes. The value $n = 3$ adopted by the glaciological community is effectively an average of $n = 4.0$ (dislocation creep) and $n = 1.8$ (superplastic flow), weighted by the relative contributions of the two mechanisms in the stress range of Glen's experiments.

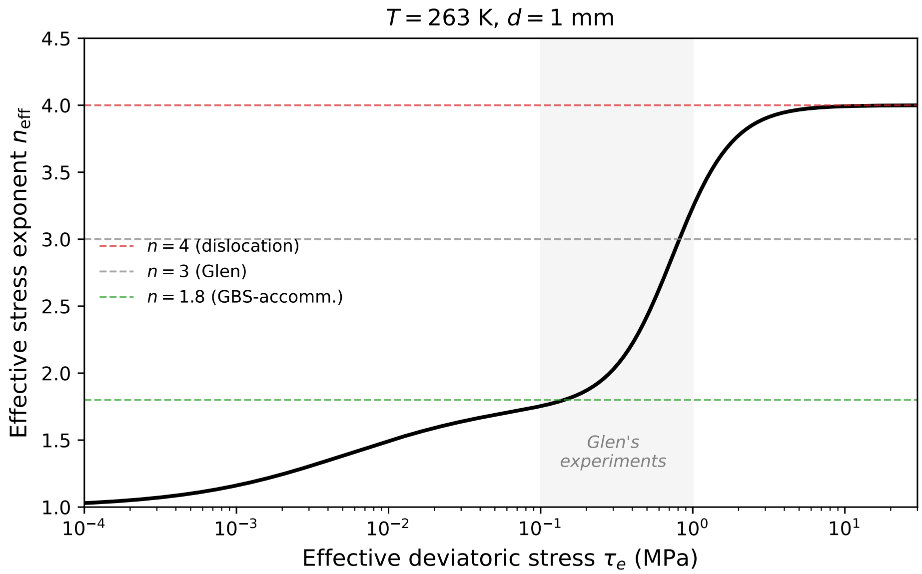

Figure 4: Effective stress exponent $n_{\mathrm{eff}} = \partial \log \dot{\varepsilon}_e / \partial \log \tau_e$ as a function of stress for $T = 263$ K, $d = 1$ mm. The exponent transitions smoothly from $n \approx 1.8$ at low stresses to $n \approx 4$ at high stresses, passing through $n \approx 3$ in Glen's experimental stress range (shaded).

Derivation: $n \approx 3$ as a transitional value [8]

Consider the composite flow law in the regime where only dislocation creep and superplastic flow contribute significantly (neglecting diffusional flow and noting that basal-slip-accommodated GBS is relevant only for very fine grains):

$$\dot{\varepsilon}_e \approx \dot{\varepsilon}_{\mathrm{gbs}} + \dot{\varepsilon}_{\mathrm{disl}} = A_{\mathrm{gbs}} \frac{\tau_e^{1.8}}{d^{1.4}} \exp\left(-\frac{Q_{\mathrm{gbs}}}{RT}\right) + A_{\mathrm{disl}}\, \tau_e^{4.0} \exp\left(-\frac{Q_{\mathrm{disl}}}{RT}\right). \tag{8}$$

At stresses $\tau_e \ll \tau_e^*$, the GBS term dominates and $\dot{\varepsilon}_e \propto \tau_e^{1.8}$. At stresses $\tau_e \gg \tau_e^*$, dislocation creep dominates and $\dot{\varepsilon}_e \propto \tau_e^{4.0}$. The crossover stress τ_e^* is the stress at which the two mechanisms contribute equally.

If one fits a single power law $\dot{\varepsilon}_e = B\, \tau_e^{n_{\mathrm{eff}}}$ over a stress range that straddles τ_e^*, the effective exponent n_{eff} varies continuously from 1.8 to 4.0, passing through $\sim$3 near the crossover. This is the origin of Glen's $n \approx 3$.

The local slope of the $\log \dot{\varepsilon}_e$–$\log \tau_e$ curve defines the apparent stress exponent at any given stress:

$$n_{\mathrm{eff}}(\tau_e) = \frac{\partial \log \dot{\varepsilon}_e}{\partial \log \tau_e} = \frac{1.8\, \dot{\varepsilon}_{\mathrm{gbs}} + 4.0\, \dot{\varepsilon}_{\mathrm{disl}}}{\dot{\varepsilon}_{\mathrm{gbs}} + \dot{\varepsilon}_{\mathrm{disl}}}. \tag{9}$$

This is a weighted average of the two stress exponents, with weights proportional to the strain-rate contribution of each mechanism.

6.2 Conditions Under Which Glen's Law Holds

Glen's law with $n = 3$ and a constant A is a reasonable approximation when:

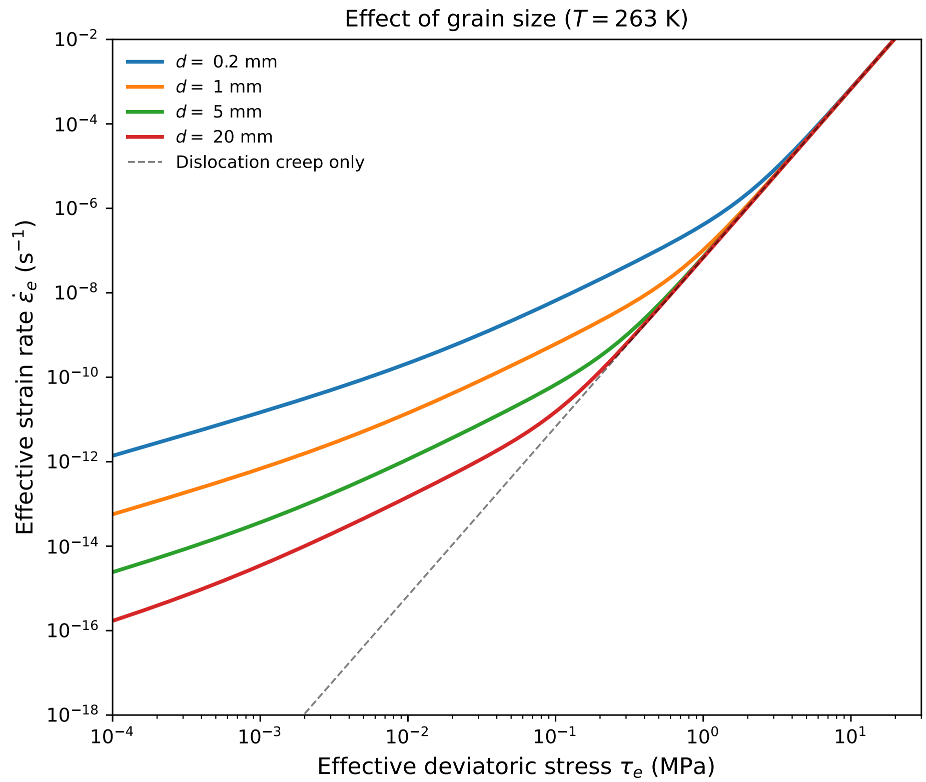

Figure 5: Composite flow law for different grain sizes at $T = 263$ K. At high stresses, all curves converge to the grain-size-independent dislocation creep line (dashed). At low stresses, finer-grained ice deforms faster due to the grain-size sensitivity of GBS-accommodated basal slip.

1. **The stress range is narrow.** Over a limited range of stresses, the curvature of the $\log \dot{\varepsilon}_e$–$\log \tau_e$ relation is small enough that a single power law provides an adequate fit.

2. **The grain size is approximately constant.** The grain-size dependence of the superplastic flow term ($r = 1.4$) is absorbed into the rate factor A. If the grain size varies spatially, a single value of A cannot capture this variation.

3. **The temperature field is known or parameterized.** The strong temperature dependence of A is typically handled separately through the Arrhenius relation.

Under these conditions, we'll use the simple form of Glen's flow law throughout this course with the understanding that this is a choice of convenience and consistency with the literature. The rate factor A is treated as a function of temperature (and implicitly of grain size, impurity content, fabric, and water content) following an Arrhenius relation:

$$A = A_0 \exp\left(-\frac{Q}{RT}\right), \tag{10}$$

with $Q \approx 60$ kJ mol^{-1} below 263 K and $Q \approx 115$–139 kJ mol^{-1} above 263 K.

6.3 What Glen's Law Does Not Capture

The reduction from the composite flow law to Glen's law discards several features that may be important in some glaciological settings:

1. **Grain-size dependence.** Glen's law is independent of grain size ($r = 0$), but the superplastic flow mechanism has $r = 1.4$. In ice with fine grains (e.g., ice-age ice with high dust content that inhibits grain growth), the effective viscosity may be lower than Glen's law predicts.

2. **Stress-dependent exponent.** Glen's law assumes a constant n, but the composite flow law predicts that the effective stress exponent varies from $\sim$1.8 at low stresses to $\sim$4.0 at high stresses. At stresses typical of ice-sheet interiors ($\tau_e \lesssim 0.01$ MPa), the effective exponent may be closer to 1.8 than to 3.

3. **Distinct activation energies.** Each creep mechanism has its own activation energy, and the effective activation energy of the composite law varies with stress. Glen's law uses a single Q (or at most two values, above and below a threshold temperature), which is an approximation.

Key point: implications for glacier modeling [8, 6, 7]

At shear stresses typical of glacier interiors ($\tau \lesssim 0.1$ MPa) and for grain sizes of 1 mm or larger, the composite flow law of [1] predicts that superplastic flow ($n = 1.8$) is the rate-limiting creep mechanism. [8] showed that dislocation creep ($n = 4$) dominates in fast-flowing regions of the Antarctic Ice Sheet, where stresses are highest, while grain-boundary sliding ($n = 1.8$) dominates elsewhere. The effective value of n thus varies spatially and temporally across an ice sheet, and the canonical $n = 3$ is an adequate representation only in the narrow transitional stress range between these two regimes.

The choice of n has direct consequences for projections of sea-level rise. [6] found that increasing n from 3 to 4 in simulations of the Amundsen Sea Embayment accelerates projected ice loss by 32±14%, with the largest sensitivity near grounding lines and shear margins where stresses are highest. [7] showed that in standard benchmark experiments, incorrectly assuming $n = 3$ when the ice behaves as $n = 4$ leads to underestimates of 100-year sea-level rise contributions of 21–35%, depending on the climate forcing. These results demonstrate that the stress exponent is not merely a theoretical nuance but a first-order control on ice-sheet projections.

Reading

Cuffey & Paterson, *Physics of Glaciers*, 4th ed.:

- Section 3.4: The flow law—overview and experimental basis
- Section 3.4.4: Creep mechanisms and the stress exponent
- Section 3.4.6: Temperature dependence and the rate factor

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